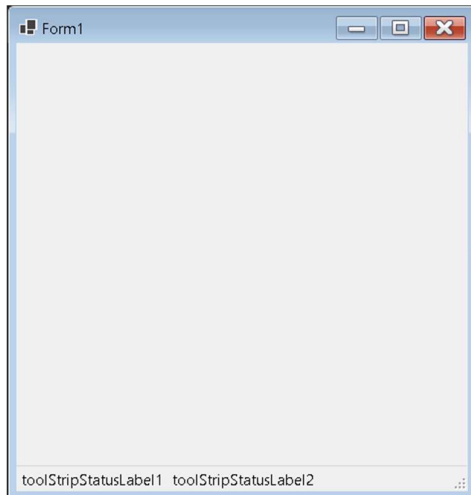


EJERCICIO PRACTICO DESARROLLO DE UN JUEGO BÁSICO

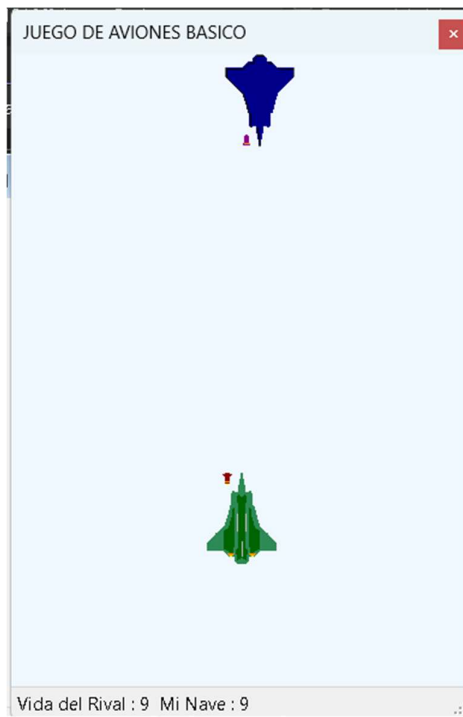
Componentes del formulario

```
Form1 System.Windows.Forms.Form
statusStrip1 System.Windows.Forms.StatusStrip
toolStripStatusLabel1 System.Windows.Forms.ToolStripStatusLabel
toolStripStatusLabel2 System.Windows.Forms.ToolStripStatusLabel
```

Vista del formulario



Vista en tiempo de ejecución



```
1
2 // *****//
3 // NOMBRE PROYECTO: JUEGO DE AVIONES BASICO //
4 // AUTOR: ING. JUAN CARLOS PINTO L. //
5 // INSTITUCION: UNAJ //
6 // FECHA DE CREACION: 01/07/2024 //
7 // VERSION : 2.0.0.0 //
8 //*****//
9
10 using System.Drawing.Drawing2D;
11
12 namespace Juego_Basic
13 {
14     public partial class Form1 : Form
15     {
16         //***** VARIABLES GLOBALES *****//
17         PictureBox navex = new PictureBox();
18         PictureBox naveRival = new PictureBox();
19         PictureBox contiene = new PictureBox();
20         System.Windows.Forms.Timer tiempo;
21         int Dispara = 0;
22         bool flag = false;
23         float angulo = 0;
24
25
26         //***** DIAGRAMAR DEL MISIL *****//
27         public void CrearMisil(int AngRotar, Color pintar, string nombre,
28             int x, int y)
29         {
30             dynamic Balas = new PictureBox();
31             int PosX = 1;
32             int PosY = 1;
33             int largoM = 11;
34             int anchoM = 8;
35             //declarar los array de puntos.
36             Point[] myMisil1 = { new Point(4 * PosX, 0 * PosY), new Point
37                 (5 * PosX, 1 * PosY), new Point(6 * PosX, 2 * PosY), new
38                 Point(6 * PosX, 7 * PosY), new Point(7 * PosX, 8 * PosY),
39                 new Point(8 * PosX, 9 * PosY), new Point(7 * PosX, 9 *
40                 PosY), new Point(6 * PosX, 10 * PosY), new Point(2 * PosX,
41                 10 * PosY), new Point(1 * PosX, 9 * PosY), new Point(0 *
42                 PosX, 9 * PosY), new Point(1 * PosX, 8 * PosY), new Point(2
43                 * PosX, 7 * PosY), new Point(2 * PosX, 2 * PosY), new Point
44                 (3 * PosX, 1 * PosY), new Point(4 * PosX, 0 * PosY) };
45             Point[] myMisil = new Point[myMisil1.Count()];
46             for (int i = 0; i < myMisil1.Count(); i++)
47             {
48                 myMisil[i].X = myMisil1[i].X;
49                 if (AngRotar == 180)
```

```

41         myMisil[i].Y = largoM - myMisil1[i].Y;
42     else
43         myMisil[i].Y = myMisil1[i].Y;
44     }
45     GraphicsPath ObjGrafico = new GraphicsPath();
46     ObjGrafico.AddPolygon(myMisil);
47     Balas.Location = new Point(x, y);
48     Balas.BackColor = pintar;
49     Balas.Size = new Size(anchoM * PosX, largoM * PosY);
50     Balas.Region = new Region(ObjGrafico);
51     contiene.Controls.Add(Balas);
52     Balas.Visible = true;
53     Balas.Tag = nombre;
54     //***** DIBUJAR COLORES *****//
55     Bitmap flag = new Bitmap(anchoM, largoM);
56     Graphics flagImagen = Graphics.FromImage(flag);
57     flagImagen.FillRectangle(Brushes.Orange, 2, 8, 5, 1);
58     flagImagen.FillRectangle(Brushes.Yellow, 3, 10, 3, 1);
59     Balas.Image = flag;
60 }
61 //*****DESCRUCTOR DEL MISIL*****//
62 private void ImpactarTick(object sender, EventArgs e)
63 {
64     // VARIABLES LOCALES
65     int X = naveRival.Location.X;
66     int Y = naveRival.Location.Y;
67     int W = naveRival.Width;
68     int H = naveRival.Height;
69     int PH = 6;
70     int X2 = navex.Location.X;
71     int Y2 = navex.Location.Y;
72     int W2 = navex.Width;
73     int H2 = navex.Height;
74     int x = naveRival.Location.X;
75     int y = naveRival.Location.Y;
76
77     Dispara++;
78     // ACCION DE DISPARAR DEL RIVAL
79     if (Dispara == 100 && naveRival.Visible == true)
80     {
81         int xRival = naveRival.Location.X + (naveRival.Width / 2);
82         int yRival = naveRival.Location.Y + (naveRival.Height / 2);
83         CrearMisil(180, Color.DarkRed, "Rival", xRival, yRival);
84         Dispara = 0;
85     }
86     // MOVIMIENTO DE LA NAVE A DESTRUIR
87     if (flag == false)
88     {

```

```
89         if (contiene.Width == x + naveRival.Width)
90             flag = true;
91         x++;
92     }
93     else
94     {
95         if (contiene.Location.X == x)
96             flag = false;
97         x--;
98     }
99     naveRival.Location = new Point(x, y);
100     naveRival.BorderStyle = BorderStyle.FixedSingle;
101     // ELIMINACION DEL MISIL Y DESCONTAR PUNTOS DE IMPACTO DE LA  ↗
102     NAVE RIVAL
103     foreach (Control c in contiene.Controls)
104     {
105         if (c is PictureBox)
106         {
107             int X1 = ((PictureBox)c).Location.X;
108             int Y1 = ((PictureBox)c).Location.Y;
109             int W1 = ((PictureBox)c).Width;
110             int H1 = ((PictureBox)c).Height;
111             string nombre = ((PictureBox)c).Tag.ToString();
112             // ACTIVIDAD DE IMPACTO CON LA NAVE RIVAL
113             if (X < X1 && X1 + W1 < X + W && Y < Y1 && Y1 + H1 <  ↗
114                 Y + H && nombre == "Misil")
115             {
116                 if (X + PH < X1 && X1 + W1 < X + W - PH)
117                 {
118                     ((PictureBox)c).Dispose();
119                     naveRival.Tag = int.Parse
120                         (naveRival.Tag.ToString()) - 10;
121                 }
122                 else
123                 {
124                     ((PictureBox)c).Dispose();
125                     naveRival.Tag = int.Parse
126                         (naveRival.Tag.ToString()) - 1;
127                 }
128                 toolStripStatusLabel1.Text = "Vida del  ↗
129                 Rival : " + naveRival.Tag.ToString();
130                 //tiempo.Stop();
131             }
132         }
133     }
134     else if (int.Parse(naveRival.Tag.ToString()) <= 0)
135     {
136         naveRival.Dispose();
137         Bitmap NuevoImg = new Bitmap(contiene.Width,  ↗
138             contiene.Height);
139         Graphics flagImagen = Graphics.FromImage  ↗
```

```
(NuevoImg);
132 // Crear cadena para dibujar.
133 String drawString = "Felicitaciones Ganaste !";
134 // Crear la fuente y el pincel.
135 Font drawFont = new Font("Arial", 16);
136 SolidBrush drawBrush = new SolidBrush(Color.Blue);
137 Point drawPoint = new Point(40, 150);
138 // Dibujar cadena en pantalla.
139 flagImagen.DrawString(drawString, drawFont,
    drawBrush, drawPoint);
140 contiene.Image = NuevoImg;
141 tiempo.Stop();
142 // Escribir resultado en archivo para pruebas
    automatizadas
143 System.IO.File.WriteAllText("resultado.txt",
    "Ganaste");
144 }
145 // ACTIVIDAD DE IMPACTO CON MI NAVE
146 if (X2 < X1 && X1 + W1 < X2 + W2 && Y2 < Y1 && Y1 +
    H1 < Y2 + H2 && nombre == "Rival")
147 {
148     if (X2 + PH < X1 && X1 + W1 < X2 + W2 - PH)
149     {
150         ((PictureBox)c).Dispose();
151         navex.Tag = int.Parse(navex.Tag.ToString()) -
            10;
152     }
153     else
154     {
155         ((PictureBox)c).Dispose();
156         navex.Tag = int.Parse(navex.Tag.ToString()) -
            1;
157     }
158     toolStripStatusLabel2.Text = "Mi Nave : " +
        navex.Tag.ToString();
159 }
160 else if (int.Parse(navex.Tag.ToString()) <= 0)
161 {
162     navex.Dispose();
163     Bitmap NuevoImg = new Bitmap(contiene.Width,
        contiene.Height);
164     Graphics flagImagen = Graphics.FromImage
        (NuevoImg);
165     // Crear cadena para dibujar.
166     String drawString = "Perdiste el Juego";
167     // Crear la fuente y el pincel.
168     Font drawFont = new Font("Arial", 16);
169     SolidBrush drawBrush = new SolidBrush(Color.Red);
170     Point drawPoint = new Point(70, 150);
```

```

171         // Dibujar cadena en pantalla.
172         flagImagen.DrawString(drawString, drawFont,
                                drawBrush, drawPoint);
173         contiene.Image = NuevoImg;
174         tiempo.Stop();
175         // Escribir resultado en archivo para pruebas
            automatizadas
176         System.IO.File.WriteAllText("resultado.txt",
            "Perdiste");

177     }
178
179     if (((PictureBox)c).Location.Y <= 0 && nombre ==
        "Misil")
180     {
181         ((PictureBox)c).Dispose();
182     }
183     if (((PictureBox)c).Location.Y >= contiene.Height &&
        nombre == "Rival")
184     {
185         ((PictureBox)c).Dispose();
186     }
187     if (nombre == "Misil")
188     {
189         ((PictureBox)c).Top -= 10;
190     }
191     if (nombre == "Rival")
192     {
193         ((PictureBox)c).Top += 10;
194     }
195     if (W >= X2 && H >= Y2 && W2 >= X && H2 >= Y)
196     {
197         naveRival.Dispose();
198         navex.Dispose();
199     }
200     }
201     else
202         tiempo.Stop();
203     }
204 }
205
206 //***** DIAGRAMAR NAVE*****//
207 public void CrearNave(PictureBox Avion, int AngRotar, int Tipox,
    Color Pintar, int Vida)
208 {
209     int largoN = 0;
210     int anchoN = 0;
211     Point[] myNave1 = { new Point(29, 0), new Point(30, 1), new
        Point(30, 6), new Point(31, 6), new Point(31, 11), new Point
        (32, 11), new Point(32, 17), new Point(35, 17), new Point

```

```

        (35, 16), new Point(37, 16), new Point(37, 17), new Point
        (38, 18), new Point(38, 28), new Point(39, 28), new Point
        (42, 39), new Point(44, 45), new Point(50, 51), new Point
        (51, 51), new Point(51, 52), new Point(58, 59), new Point
        (58, 66), new Point(44, 66), new Point(39, 71), new Point
        (35, 71), new Point(35, 74), new Point(32, 77), new Point
        (26, 77), new Point(23, 74), new Point(23, 71), new Point
        (19, 71), new Point(14, 66), new Point(0, 66), new Point(0,
        59), new Point(7, 52), new Point(7, 51), new Point(8, 51),
        new Point(14, 45), new Point(16, 39), new Point(19, 28),
        new Point(20, 28), new Point(20, 18), new Point(21, 17),
        new Point(21, 16), new Point(23, 16), new Point(23, 17),
        new Point(26, 17), new Point(26, 11), new Point(27, 11),
        new Point(27, 6), new Point(28, 6), new Point(28, 1), new
        Point(29, 0) };
212 Point[] myNave2 = { new Point(24, 0), new Point(29, 5), new
        Point(29, 18), new Point(32, 21), new Point(34, 21), new
        Point(38, 17), new Point(41, 20), new Point(41, 30), new
        Point(47, 36), new Point(47, 41), new Point(41, 41), new
        Point(38, 44), new Point(36, 44), new Point(33, 41), new
        Point(30, 41), new Point(25, 46), new Point(22, 46), new
        Point(17, 41), new Point(14, 41), new Point(11, 44), new
        Point(9, 44), new Point(6, 41), new Point(0, 41), new Point
        (0, 36), new Point(6, 30), new Point(6, 20), new Point(9,
        17), new Point(13, 21), new Point(15, 21), new Point(18,
        18), new Point(18, 5), new Point(23, 0) };
213 Point[] myNave3 = { new Point(25, 54), new Point(26, 54), new
        Point(26, 50), new Point(26, 50), new Point(27, 50), new
        Point(28, 50), new Point(29, 50), new Point(30, 51), new
        Point(31, 51), new Point(32, 52), new Point(32, 49), new
        Point(31, 48), new Point(30, 47), new Point(29, 46), new
        Point(28, 45), new Point(27, 44), new Point(27, 36), new
        Point(28, 35), new Point(28, 25), new Point(29, 25), new
        Point(30, 25), new Point(31, 25), new Point(32, 26), new
        Point(33, 26), new Point(34, 27), new Point(35, 28), new
        Point(36, 28), new Point(37, 29), new Point(38, 30), new
        Point(39, 30), new Point(40, 31), new Point(41, 32), new
        Point(42, 32), new Point(43, 33), new Point(44, 34), new
        Point(45, 35), new Point(46, 36), new Point(47, 36), new
        Point(48, 36), new Point(49, 37), new Point(50, 37), new
        Point(51, 38), new Point(51, 37), new Point(51, 36), new
        Point(51, 35), new Point(50, 35), new Point(37, 22), new
        Point(37, 15), new Point(36, 14), new Point(35, 14), new
        Point(34, 15), new Point(34, 21), new Point(28, 15), new
        Point(28, 7), new Point(27, 6), new Point(26, 5), new Point
        (25, 5), new Point(24, 6), new Point(23, 7), new Point(23,
        15), new Point(17, 21), new Point(17, 15), new Point(16,
        14), new Point(15, 14), new Point(14, 15), new Point(14,
        22), new Point(1, 35), new Point(0, 35), new Point(0, 36),

```

```

        new Point(0, 37), new Point(0, 38), new Point(1, 37), new
        Point(2, 37), new Point(3, 36), new Point(4, 36), new Point
        (5, 36), new Point(6, 35), new Point(7, 34), new Point(8,
        33), new Point(9, 32), new Point(10, 32), new Point(11,
        31), new Point(12, 30), new Point(13, 30), new Point(14,
        29), new Point(15, 28), new Point(16, 28), new Point(17,
        27), new Point(18, 26), new Point(19, 26), new Point(20,
        25), new Point(21, 25), new Point(22, 25), new Point(23,
        25), new Point(23, 35), new Point(24, 36), new Point(24,
        44), new Point(23, 45), new Point(22, 46), new Point(21,
        47), new Point(20, 48), new Point(19, 49), new Point(19,
        52), new Point(20, 51), new Point(21, 51), new Point(22,
        50), new Point(23, 50), new Point(24, 50), new Point(25,
        50), new Point(25, 54) };
214 Point[] myNave;
215 //*****INSERTAR EL OBJETO*****//
216 GraphicsPath ObjGrafico = new GraphicsPath();
217 if (Tipox == 1)
218 {
219     largoN = 77;
220     anchoN = 58;
221     // rotar
222     myNave = new Point[myNave1.Count()];
223     for (int i = 0; i < myNave1.Count(); i++)
224     {
225         myNave[i].X = myNave1[i].X;
226         if (AngRotar == 180)
227             myNave[i].Y = largoN - myNave1[i].Y;
228         else
229             myNave[i].Y = myNave1[i].Y;
230     }
231     ObjGrafico.AddPolygon(myNave);
232 }
233 else if (Tipox == 2)
234 {
235     largoN = 46 * 1;
236     anchoN = 47 * 1;
237     // rotar
238     myNave = new Point[myNave2.Count()];
239     for (int i = 0; i < myNave2.Count(); i++)
240     {
241         myNave[i].X = myNave2[i].X * 1;
242         if (AngRotar == 180)
243             myNave[i].Y = largoN - myNave2[i].Y * 1;
244         else
245             myNave[i].Y = myNave2[i].Y * 1;
246     }
247     //ObjGrafico.AddPolygon(myNave);
248     ObjGrafico.AddLines(myNave);

```

```

249     }
250     else if (Tipox == 3)
251     {
252         largoN = 54;
253         anchoN = 51; // rotar
254         myNave = new Point[myNave3.Count()];
255         for (int i = 0; i < myNave3.Count(); i++)
256         {
257             myNave[i].X = myNave3[i].X;
258             if (AngRotar == 180)
259                 myNave[i].Y = largoN - myNave3[i].Y;
260             else
261                 myNave[i].Y = myNave3[i].Y;
262         }
263         ObjGrafico.AddPolygon(myNave);
264     }
265     Avion.BackColor = Pintar;
266     Avion.Size = new Size(anchoN, largoN);
267
268     Avion.Region = new Region(ObjGrafico);
269     Avion.Location = new Point(0, 0);
270     //*****INSERTAR LA NAVE AL CONTENDOR*****//
271     contiene.Controls.Add(Avion);
272
273     Bitmap Imagen = new Bitmap(Avion.Width, Avion.Height);
274     Graphics PintaImg = Graphics.FromImage(Imagen);
275     Point[] Colorea = { new Point(24, 2), new Point(27, 5), new Point(27, 18), new Point(31, 22), new Point(34, 22), new Point(37, 19), new Point(38, 19), new Point(39, 20), new Point(39, 30), new Point(45, 36), new Point(45, 39), new Point(41, 39), new Point(38, 42), new Point(35, 42), new Point(32, 39), new Point(30, 39), new Point(25, 44), new Point(21, 44), new Point(16, 39), new Point(14, 39), new Point(11, 42), new Point(8, 42), new Point(5, 39), new Point(1, 39), new Point(1, 36), new Point(7, 30), new Point(7, 20), new Point(8, 19), new Point(9, 19), new Point(12, 22), new Point(15, 22), new Point(19, 18), new Point(19, 5), new Point(22, 2) };
276
277     //PintaImg.FillPolygon(Brushes.DarkGreen, Colorea);
278     Point[] poix = new Point[myNave2.Count()];
279     for (int i = 0; i < myNave2.Count(); i++)
280     {
281         poix[i].X = myNave2[i].X;
282         poix[i].Y = myNave2[i].Y;
283     }
284     PintaImg.DrawPolygon(Pens.Black, ObjGrafico.PathData.Points);
285     Avion.Image = Imagen;
286

```

```

287         //NaveCorre(Avion,Tipox,AngRotar);
288         Avion.Tag = Vida;
289         Avion.Visible = true;
290     }
291
292     //***** EFECTOS DE LA NAVE PRINCIPAL *****//
293     public void NaveCorre(PictureBox Avion, int AngRotar, int velox)
294     {
295         Bitmap Imagen = new Bitmap(Avion.Width, Avion.Height);
296         Graphics PintaImg = Graphics.FromImage(Imagen);
297         Point[] puntoDer = { new Point(35, 28), new Point(35, 30),
298                             new Point(36, 30), new Point(37, 31), new Point(37, 37),
299                             new Point(38, 38), new Point(38, 40), new Point(39, 41),
300                             new Point(39, 44), new Point(40, 45), new Point(40, 46),
301                             new Point(42, 48), new Point(43, 48), new Point(44, 49),
302                             new Point(44, 64), new Point(43, 65), new Point(42, 65),
303                             new Point(41, 66), new Point(40, 66), new Point(38, 68),
304                             new Point(36, 68), new Point(36, 69), new Point(36, 63),
305                             new Point(35, 62), new Point(35, 28) };
306         Point[] puntoIzq = { new Point(23, 28), new Point(23, 30),
307                             new Point(22, 30), new Point(21, 31), new Point(21, 37),
308                             new Point(20, 38), new Point(20, 40), new Point(19, 41),
309                             new Point(19, 44), new Point(18, 45), new Point(18, 46),
310                             new Point(16, 48), new Point(15, 48), new Point(14, 49),
311                             new Point(14, 64), new Point(15, 65), new Point(16, 65),
312                             new Point(17, 66), new Point(18, 66), new Point(20, 68),
313                             new Point(22, 68), new Point(22, 69), new Point(22, 63),
314                             new Point(23, 62), new Point(23, 28) };
315         Point[] puntoAtr = { new Point(29, 21), new Point(31, 19),
316                             new Point(32, 19), new Point(33, 20), new Point(33, 25),
317                             new Point(32, 26), new Point(32, 63), new Point(34, 65),
318                             new Point(34, 68), new Point(33, 69), new Point(33, 74),
319                             new Point(32, 73), new Point(31, 73), new Point(29, 71),
320                             new Point(27, 73), new Point(26, 73), new Point(25, 74),
321                             new Point(25, 69), new Point(24, 68), new Point(24, 65),
322                             new Point(26, 63), new Point(26, 26), new Point(25, 25),
323                             new Point(25, 20), new Point(26, 19), new Point(27, 19),
324                             new Point(29, 21) };
325
326         PintaImg.FillPolygon(Brushes.DarkGreen, puntoDer);
327         PintaImg.FillPolygon(Brushes.DarkGreen, puntoIzq);
328         PintaImg.FillPolygon(Brushes.DarkGreen, puntoAtr);
329         PintaImg.FillRectangle(Brushes.Silver, 25, 35, 1, 15);
330         PintaImg.FillRectangle(Brushes.Silver, 32, 35, 1, 15);
331         PintaImg.FillRectangle(Brushes.Silver, 29, 58, 1, 13);
332         if (velox == 1)
333         {
334             PintaImg.FillRectangle(Brushes.DarkOrange, 35, 68, 6, 1);
335             PintaImg.FillRectangle(Brushes.Orange, 36, 69, 4, 1);
336         }
337     }

```

```
311      PintaImg.FillRectangle(Brushes.Yellow, 37, 70, 2, 1);
312      PintaImg.FillRectangle(Brushes.DarkOrange, 17, 68, 6, 1);
313      PintaImg.FillRectangle(Brushes.Orange, 18, 69, 4, 1);
314      PintaImg.FillRectangle(Brushes.Yellow, 19, 70, 2, 1);
315      RotateImage(navex.Image, angulo);
316  }
317  else if (velox == 2)
318  {
319      PintaImg.FillRectangle(Brushes.DarkRed, 15, 30, 1, 8);
320      PintaImg.FillRectangle(Brushes.DarkRed, 25, 28, 1, 16);
321      PintaImg.FillRectangle(Brushes.DarkRed, 35, 30, 1, 8);
322  }
323  else if (velox == 3)
324  {
325      PintaImg.FillRectangle(Brushes.DarkRed, 15, 30, 1, 8);
326      PintaImg.FillRectangle(Brushes.DarkRed, 25, 28, 1, 16);
327      PintaImg.FillRectangle(Brushes.DarkRed, 35, 30, 1, 8);
328  }
329  Avion.Image = RotateImage(Imagen, AngRotar);
330  }
331
332  //*****CREAR ANGULO DE ROTACION*****//
333  public static Image RotateImage(Image img, float rotationAngle)
334  {
335      Bitmap bmp = new Bitmap(img.Width, img.Height);
336      Graphics gfx = Graphics.FromImage(bmp);
337      gfx.TranslateTransform((float)bmp.Width / 2, (float)      ↗
338          bmp.Height / 2);
339      gfx.RotateTransform(rotationAngle);
340      gfx.TranslateTransform(-(float)bmp.Width / 2, -(float)      ↗
341          bmp.Height / 2);
342      gfx.InterpolationMode = InterpolationMode.HighQualityBicubic;
343      gfx.DrawImage(img, new Point(0, 0));
344      gfx.Dispose();
345      return bmp;
346  }
347
348  //***** MOVIMIENTO DEL TECLADO DEL USUARIO *****//
349  public void ActividadTecla(object sender, KeyEventArgs e)
350  {
351      //INSTRUCCIONES DE LOS BOTONES
352      switch (e.KeyValue)
353      {
354          case 37: // flecha hacia la izquierda
355              {
356                  if (contiene.Left < navex.Left) // PARAMETROS DE      ↗
357                      LIMITE
358                      navex.Left -= 10;
359                      angulo = -15;
```

```
357         NaveCorre(navex, 1, 0);
358         break;
359     }
360     case 38: // flecha hacia arriba
361     {
362         if (contiene.Top < navex.Top)
363             navex.Top -= 10;
364         NaveCorre(navex, 0, 1);
365         break;
366     }
367     case 39: // flecha hacia la derecha
368     {
369         if (contiene.Right > navex.Right)
370             navex.Left += 10;
371         angulo = +15;
372         NaveCorre(navex, 1, 0);
373         break;
374     }
375     case 40: // flecha hacia abajo
376     {
377         if (contiene.Bottom > navex.Bottom)
378             navex.Top += 10;
379         NaveCorre(navex, 0, 1);
380         break;
381     }
382     case 13:
383     {
384         tiempo.Start();
385         int x = navex.Location.X + (navex.Width / 2);
386         int y = navex.Location.Y + (navex.Height / 2);
387         CrearMisil(0, Color.DarkMagenta, "Misil", x, y);
388         break;
389     }
390 }
391 }
392
393 //***** ACTIVAR ACCIONES DE INICIALIZACION *****
394 public void Iniciar()
395 {
396     //*****
397     this.FormBorderStyle = FormBorderStyle.SizableToolWindow;
398     this.Width = 400;
399     this.Height = 600;
400     this.Text = "JUEGO DE AVIONES BASICO";
401     toolStripStatusLabel1.Text = "Mi Rival";
402     toolStripStatusLabel2.Text = "Mi Avion";
403     this.KeyDown += new KeyEventHandler(ActividadTecla);
404     //*****
```

```
405         contiene.Location = new Point(0, 0);
406         contiene.BackColor = Color.AliceBlue;
407         contiene.Size = new Size(400, 600);
408         contiene.Dock = DockStyle.Fill;
409         Controls.Add(contiene);
410         contiene.Visible = true;
411         //-----//
412         // contenido del formulario
413         Random r = new Random();
414         int aleatY = r.Next(250, 330);
415         int aleatX = r.Next(50, 250);
416         CrearNave(navex, 0, 1, Color.SeaGreen, 20);
417         //ELEGIR NAVE DE SALIDA RIVAL
418         Random sal = new Random();
419         int sale = sal.Next(1, 3);
420         CrearNave(naveRival, 180, sale, Color.DarkBlue, 50);
421         //Modulo.Escenario(contiene, sale);
422         navex.Location = new Point(aleatX, aleatY);
423
424         tiempo = new System.Windows.Forms.Timer();
425         tiempo.Interval = 5;
426         tiempo.Enabled = true;
427         tiempo.Tick += new EventHandler(ImpactarTick);
428
429         this.KeyPreview = true;
430     }
431
432
433
434     public Form1()
435     {
436         InitializeComponent();
437         Iniciar();
438     }
439 }
440 }
441
```